

DATASET INSPECTION REPORT: TELCO CUSTOMER CHURN

Prepared By- Ojo Precious Mojolaoluwa

1. Dataset Overview

The dataset contains customer information from a telecommunications company, aimed at predicting customer churn.

- **Total Features:** 22 columns (1 Identifier, 19 Predictors, 2 Target representations).
- **Target Variables:** Churn (Categorical: Yes/No) and Churn_Flag (Binary: 0/1).
- **Target Audience:** Data Science / Analytics teams for binary classification modeling.

2. Feature Categorization

The variables can be logically grouped into four main categories:

- **Demographic Information:** gender, SeniorCitizen (0/1), Partner (Yes/No), Dependents (Yes/No).
- **Account & Financial Attributes:** tenure (months with the company), Contract (Month-to-month, One year, Two year), PaperlessBilling (Yes/No), PaymentMethod, MonthlyCharges, TotalCharges.
- **Service Subscriptions:**
 - *Primary:* PhoneService, InternetService (DSL, Fiber optic, No).
 - *Add-ons:* MultipleLines, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies.
- **Identifier:** customerID (Unique alphanumeric string).

3. Data Types & Value Domains

- **Numerical (Continuous/Discrete):** tenure, MonthlyCharges, TotalCharges.
- **Binary/Categorical:** SeniorCitizen, Churn_Flag (0/1).
- **Categorical (Multi-class):** Contract, PaymentMethod, InternetService.
- **Categorical (Conditional):** Service add-ons contain three distinct values: *Yes*, *No*, and *No internet/phone service* (indicating logical dependency on primary services).

4. Key Data Quality Observations

- **Missing Values in TotalCharges:** An inspection of the raw data reveals blank/missing values in the TotalCharges column. Cross-referencing shows these blanks exclusively

occur where tenure is 0. These represent newly acquired customers who have not yet completed their first billing cycle.

- **Redundant Target Columns:** The dataset includes both the raw text target (Churn) and the encoded target (Churn_Flag).
- **Logical Hierarchies:** Features like MultipleLines depend on PhoneService, and streaming/security features depend on InternetService. The dataset handles this gracefully by using the "No internet/phone service" category rather than leaving them as nulls.

5. Recommendations for Preprocessing

1. **Handle Missing Financials:** Replace the blank spaces in TotalCharges with 0.0 (since tenure is 0) or drop these specific rows if the focus is strictly on historical customer behavior.
2. **Drop Identifiers:** Remove customerID prior to model training as it holds no predictive weight and will cause overfitting.
3. **Drop Redundancy:** Drop the Churn text column and retain only Churn_Flag as the target variable for machine learning algorithms.
4. **Encoding:** Apply One-Hot Encoding to multi-class categorical variables (e.g., PaymentMethod, Contract, InternetService) and Label Encoding (0/1) to binary "Yes/No" features.